

Figure 7: IG1, Interference graph (after ILP)

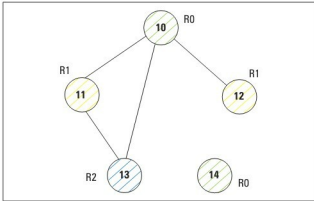


Figure 8: Interference graph after colouring

Now, we construct the interference graph for the code with ILP. We'll call this graph IG1 (shown in Figure 7).

Now, we try to colour the above graph. Let's suppose that the maximum number of available physical registers is three—namely, *R0*, *R1*, and *R2*. The coloured graph will be as shown in Figure 8.

Here, we see that the graph is colourable, i.e., we are able to colour the graph with the available three registers. Hence, this is the best possible solution for the given code.

Next, let's look at a case when the graph is not colourable. Again, let's suppose that the maximum number of available physical registers is three. Consider the following example:

```

1.  MOVE 10, [a] ;
2.  MOVE 11, [b] ;
3.  ADD 12, 10, 11 ;
4.  ADD 13, 10, 10 ;
5.  ADD 14, 10, 11 ;
6.  MOVE 15, 14 ;
7.  ADD 16, 13, 13 ;

```

To this code, we apply ILP as:

```

1.  MOVE 10, [a] ;
2.  MOVE 11, [b]      ADD 13, 10, 10 ;

```

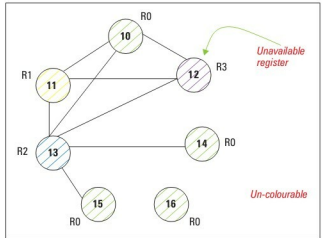


Figure 9: Un-colourable interference graph (after ILP)

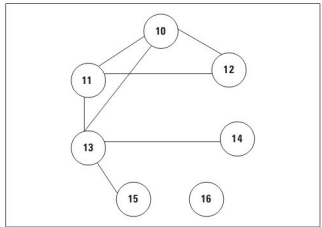


Figure 10: IG2, Interference graph (before ILP)

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3.  ADD 12, 10, 11 ;
4.  ADD 14, 10, 11 ;
5.  MOVE 15, 14 ;
6.  ADD 16, 13, 13 ;

```

Now we construct the interference graph for the resultant code, as shown in Figure 9. We see that after applying ILP, the resultant interference graph is un-colourable because to colour Node 12 in the IG, we would require a fourth register, but we only have three registers available.

Now, we return to the original code that is without ILP, and construct the interference graph IG2 for it, as shown in Figure 10. We try to colour this graph. We see that the resultant graph, shown in Figure 11, is colourable—we can colour all the nodes with the given three registers.

If we compare the two graphs (before ILP, colourable, and after ILP, un-colourable), as shown in Figure 12, we see that the interference graph after ILP has an additional edge, due to which the graph becomes un-colourable.

Now, we may have two possible solutions to deal with the un-colourable graph. The first is to ignore the edge (between